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Product:	**Laricyl® BC10206**
CAS No.:	107-88-0, 94465-74-4, 61788-85-0
INCI name:	Butylene Glycol, Fomes Officinalis Extract, PEG-40 Hydrogenated Castor Oil

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GENERAL

This evaluation refers exclusively to the product Laricyl® BC10206 in the quality as specified and provided by BASF.

The product was toxicologically evaluated on the basis of data obtained with a close homolog (same concentration of ingredients and additionally 0.8% Phenoxyethanol).

This toxicological evaluation complies with the "SCCS-Notes of guidance for testing of cosmetic ingredients for their safety evaluation" (1). Besides of screening assays, studies dated later than 1984 were conducted in full compliance with the "OECD-principles of Good Laboratory Practice"(2). Classification with respect to the individual endpoints complies with the criteria for the classification and labelling of substances and preparations in the European Union.

1 Acute oral toxicity

The acute oral toxicity was investigated with the homolog of Laricyl® BC10206 in five male und five female rats, specific pathogen free Sprague-Dawley albino rats, according to a single administration test procedure. A single dose of 2000 mg/kg body weight was administrated by gavage to animals that had previously been fasted. No adverse effects were observed during a 14-day post-application observation period. Thus, the LD_{50} exceeded by far 2000 mg/kg (3).

Based on this test, the potential of Laricyl® BC10206 to cause acute systemic toxicity can reliably be evaluated.

On the basis of this test, classification and labelling of Laricyl® BC10206 is not necessary.

2 Dermal toxicity

Experimental determination of dermal toxicity was omitted for the following analogous consideration: in general, invasion via the gastro-intestinal route renders a significantly higher bioavailability compared to the dermal route. As the substance applied orally (a worst-case condition compared to dermal application) beyond the classification limit (> 2000 mg/kg for the acute toxicity) did not cause any signs of systemic toxicity, this can be deduced also for the cutaneous route of application.

3 Dermal irritation

3.1 In-vitro testing (SKINTEX assay)

Preliminary and predictive information can be obtained from *in-vitro* screening tests using skin model matrices or natural skin specimens, which are brought into contact with various concentrations of the substance to be tested. Its dose-related impact on the model membrane is correlated with established, destructive effects induced by certain standard compounds in the same system.

The homolog of Laricyl® BC10206 was applied as 15 % dilution in the so-called SKINTEX assay which is based on an artificial collagen/keratin matrix along with an integrated colorant as indicator for measuring membrane integrity.

There was no significant dose-related activity of the test item (up to 100 µl tested) on the membrane structure: therefore, the homolog was ranked as “non-irritant” according to the used ranking system [(4)].

3.2 Primary skin irritation

3.2.1 Draize test in rabbits

The acute dermal irritation was tested according to Draize with 0.5 ml of a 15 % aqueous dilution (w/v) of the homolog of Laricyl® BC10206 in 6 rabbits, strain White New Zealand. The contact time under occlusive conditions was 24 hours. The 1- and 48-hours scores were 0 for the erythema and 0 for the oedema in all animals. No irritant effects were observed on the exposed intact and abraded skin area [(5)].

3.2.2 Human patch test

Within the scope of a patch-testing study, 52 human volunteers received repeated topical applications of 0.2 ml of the homolog of Laricyl® BC10206 (15 % aqueous dilution, w/v) within 3 weeks during the induction phase. The contact was 24 hours each time under occlusive conditions. No signs of skin irritation, except low-level reactions in three subjects during the induction phase, were noted in any individual [(10)].

In light of this data, the tested preparation up to at least a 15 % aqueous dilution can be evaluated as well skin compatible.

4 Mucous membrane irritation

4.1 *In-vitro* testing

In-vitro studies are available which serve as a preliminary screening for detecting clear signs of irritation:

4.1.1 EYTEX assay

The so-called EYTEX *in-vitro* assay was performed with the homolog of Laricyl® BC10206 (15 % aqueous dilution, up to 100 µl). A model matrix mainly composed of proteins, glycoproteins and mucopolysaccharides is employed as surrogate of the eye cornea. Dose-related coagulation and aggregation produced by the test substance is correlated with established, destructive effects induced by certain standard compounds in the same system.

There was no evidence of a significant activity, although a very slight dose-related irritant activity of the test item on the matrix structure was noted. Therefore, the homolog was ranked as “very weakly irritating” according to the used ranking system. [6].

4.1.2 HET-CAM assay (Hen’s egg test with chorioallantois membrane)

Furthermore, the homolog of Laricyl® BC10206 was tested in the so-called HET-CAM assay. The chorioallantois membrane of a chicken embryo in hen´s egg is employed as a susceptible test medium representing a natural mucous membrane [7,8]. 0.3 ml test substance (15 % aqueous dilution, w/v) was applied onto the membrane after 10-d incubation of the fertilized eggs. Following a contact time of 20 sec, the appearance of irritant effects (hyperemia, hemorrhage and coagulation) were evaluated and scored over time within 5 min post-treatment. According to the used ranking system, the homolog was classified as “slightly irritating“ due to hyperemia produced in all four test eggs after 2 - 5 min post-treatment. No other signs of irritation were observed [8].

4.2 Primary eye irritation in rabbits

Acute eye irritation was tested with the homolog of Laricyl® BC10206 in 3 albino rabbits, strain White New Zealand. 0.1 ml of a 15 % dilution (w/v) in aqua was instilled by single application and permanent contact into the right eyes of the rabbits, while the left ones served as the experimental controls. The eyes were scored 1-hour, 1-, 2-, 3-, 4- and 7-days post-application.

All three animals showed slight redness and lacrimation within 1 h after treatment. All signs of irritation had completely vanished within 48 h. The individual scores for erythema and lacrimation were not higher than 1 of maximally 3 possible scores. Changes of the iris and the cornea were not observed. The product was graded as “weakly irritating” to the eye [5].

Based on the EU classification system, the weak and transient reaction of the conjunctivae only does not require a classification of the homolog and consequently of Laricyl® BC10206 with regard to the effects to the eyes up to a 15-% aqueous dilution.

5 Skin sensitization

5.1 Maximization test in guinea pigs

The potential skin sensitizing properties of the homolog of Laricyl® BC10206 (15 % aqueous dilution for intra- and epidermal induction and 15 and 10 % for challenge) were assessed in the adjuvant-assisted Maximization test in guinea pigs, strain DUNKIN HARTLEY, according to the Magnusson-Kligman protocol. Under these experimental conditions, at 10 and 15 % dilutions of Laricyl® BC10206, no skin reactions were seen, indicating neither irritating nor allergic potential. The sensitization rate at 24 h and at 48 h was 0 % [(9)].

5.2 Human repeated insult patch test

Results from a human repeated insult patch test further support the lack of sensitizing properties of Laricyl® BC10206. A 10 % aqueous dilution (w/v) of Laricyl® BC10206was repeatedly applied (total of 9 applications within 3 weeks) for 24 h under occlusive conditions to the backs of 120 volunteers during the induction phase. The challenge took place two weeks later with the same concentration and at the same area as well as at a naive site of the back. Readings were taken 24 h after removing of the patches during induction, as well as 48, 72 and 96 h after removing of the challenge patches. Among the 105 volunteers completing the study, no skin reactions were noted during both induction and challenge [(15)].

6 Mutagenicity

Laricyl® BC10206 was tested in the Ames assay to assess the mutagenic potential in histidine dependent auxotrophic mutants of Salmonella typhimurium, strains TA1535, TA1537, TA98 and TA100, and the tryptophan dependent mutants of Escherichia coli, strain WP2uvrA. The bacteria were exposed to the test substance diluted in purified water. Purified water was also used as a negative control. Two independent mutation tests were performed in the presence and absence of liver preparations from Aroclor 1254-induced rats (S9 mix). The first (range-finding) was a Standard plate incorporation assay; the second involved a pre-incubation stage.

Concentrations up to 5000 µg/plate were tested in the mutation tests. The test substance did not induce precipitation. No signs of toxicity were observed towards the tester strains in either mutation test. No evidence of mutagenic activity was seen at any concentration of the product in either mutation test. The concurrent positive controls demonstrated the sensitivity of the assay and the metabolizing activity of the liver preparations.

It is concluded that, under the test conditions employed, Laricyl® BC10206 showed no evidence of mutagenic activity in this bacterial system [(13)].

7 Repeated dose toxicity

Experimental determination of the repeated dose toxicity was omitted based on the following considerations:
Laricyl® BC10206 consists of Butylene Glycol and small amounts of Fomes officinalis extract. Several long-term studies with Butylene Glycol in rodents and non-rodents are reported in the literature [14]. From these studies a NOAEL > 2000 mg/kg bw can be calculated. Due to the extraction process the dry extract of Fomes officinalis is based on substances with a high polarity. Based on high polarity of all compounds, skin penetration and systemic availability is expected to be rather limited. Thus, taken also the mass content of the individual ingredients into account toxic effects after prolonged repeated application have not to be expected. Furthermore, oral ingestion can be considered negligible under practical daily use conditions.

Thus, systemic toxic effects after prolonged repeated topical application have not to be expected for Laricyl® BC10206 under use conditions.

COMMENTS

The data are based on our current knowledge and are not claimed to be complete. The data do not relieve the user from the responsibility to check the suitability of the product for the intended use and from a risk assessment of its own products.

Our data may only be quoted with the consent of BASF.

References

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